

A Kali Linux virtual machine was hardened against common attack vectors using industry-standard security controls and vulnerability management practices. Nessus Essentials was used to perform both pre-hardening and post-hardening vulnerability scans to identify and validate security improvements. Prior to hardening, the system showed one Low-severity vulnerability and four Informational findings. After applying security improvements including ICMP hardening, system updates, service validation, and SSH verification, the system remained at one Low-severity finding with six Informational findings, while maintaining zero Critical, High, and Medium vulnerabilities. The project demonstrated improved security posture and reduced attack surface through documented system hardening.

Severity	Pre-Hardening Count	Post-Hardening Count
Critical	0	0
High	0	0
Medium	0	0
Low	1	1
Informational	4	6

- Restricted ICMP broadcast responses to reduce information disclosure and reconnaissance opportunities for attackers.
- Updated all Kali Linux system packages to ensure known vulnerabilities and outdated software were patched.
- Reviewed active listening services using `ss -tulnp` and confirmed only essential services were exposed.
- Verified that SSH remote access was already disabled, reducing the risk of brute-force login attempts and unauthorized remote access.
- Validated that no unnecessary legacy services such as FTP or Telnet were running, helping maintain a minimal attack surface.

The post-hardening scan still showed one Low-severity finding related to ICMP timestamp request disclosure. This finding could not be fully removed because Linux kernel handling of ICMP responses can vary depending on system configuration and Nessus detection methods. However, ICMP broadcast restrictions were successfully applied to reduce unnecessary network information disclosure.

The hardening step with the biggest impact on security was updating system packages. Keeping the operating system and installed software fully updated reduces exposure to known vulnerabilities and is one of the most important defenses against exploitation.

This process is very similar to what would be done on a real corporate workstation. Security teams often perform baseline vulnerability scans, apply hardening changes, validate services, and re-scan systems to confirm risk reduction. In a production environment, additional steps would include centralized logging, endpoint detection and response (EDR), stronger access controls, MFA enforcement, and credentialed vulnerability scans for deeper visibility.